Project-Utopia: A Tool-Grounded Multi-Channel Benchmark for LLM Hierarchical Decision-Making, Stale-Coherence Memory Failures, and Cross-Vendor Channel Routing

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Abstract

LLM-agent benchmarks today measure the wrong layer of the agent stack. AgentBench [16], GAIA [21], WebArena [38], Toolformer [26], Gorilla [23], ToolLLM [25] and MINT [34] all evaluate *tool selection*: given many tools, can the LLM pick the right one? But the failure mode that actually breaks production agent systems [3, 20, 27] is the opposite regime: the tool set is *fixed and known a priori*, and the agent must **orchestrate multiple parallel decision channels into that fixed surface coherently**. A weather update from one channel, a worker policy from another, and a build plan from a third are each individually schema-valid, yet jointly inconsistent because they reference state that has drifted between their cadences. No existing benchmark isolates this failure mode.

We introduce **Project-Utopia**, a benchmark for *multi-channel orchestration of a fixed tool surface*. The agent exposes four parallel decision channels (*environment-director*, *npc-policy*, *strategic-plan*, *colony-agent*) operating at three nested cadences over a fixed 4-tool surface (path queries over a versioned grid, Boids group steering, FSM intent application, build planning). Every directive is filtered through a per-channel pydantic schema and idempotent numerical guardrails that constitute the **agent–tool interface contract**: weights $\in [0, 3]$, TTLs $\in [8\,h, 24\,h]$, raid-posture clamps, allowlist-restricted intent and target vocabularies. The contract is the same role as ToolLLM's typed function signatures generalized to four parallel channels, plus a deterministic substrate (seeded RNG, A*, Boids, FSM) so hardware variance does not mask reasoning failures.

We evaluate the orchestration regime through two pre-registered hypotheses. **(H1)** Under matched per-decision token budget, the 4-channel orchestration contract achieves $\Delta\text{RAE} \geq 0.10$ over a flat single-channel baseline that emits per-worker FSM-state preferences directly — target effect size anchored on ChatDev's -44% role-removal effect [24]. **(H2)** Schema validity is preserved under coherence stress: across 7 200-tick runs, the fraction of directives rejected by guardrails or producing behaviorally inconsistent state remains bounded by the per-channel schema rejection rate, with no compounding inter-channel violation. The benchmark surfaces 19 dimension keys via a 4-layer metric stack (per-tick plugins → DimensionNormalizer → MeltingPot-style sandwich normalization [1] → Bayesian Beta-Binomial [5]), with PolicyValue Baseline [6] variance reduction. Reproducibility is enforced by a three-tier contract [37]: bit-identical fallback hashes verified at 30 / 1 800 / 7 200 ticks (`be19781c.../ f7a099a4.../ 43cb6aec...`).

At submission, the pipeline is end-to-end verified on fallback-only NDJSON; cell-level real-LLM signal is reserved for the camera-ready once API access is provisioned. Project-Utopia is released as a self-contained Python package (`pip install project-utopia`, 615 tests passing, $\sim$23 k LOC, 128 modules) at [ANONYMIZED-REPO-URL] (de-anonymized for camera-ready).

1 Introduction

LLM-agent benchmarks measure the wrong layer of the agent stack. The dominant evaluation regime — ReAct [36], Toolformer [26], Gorilla [23], ToolLLM [25], AgentBench [16], GAIA [21], MINT [34] — presents the agent with an *open inventory* of 10^3–10^4 tools and scores whether it selects and parameterizes the right one. Frontier models now do this competently. The mode that actually breaks deployments is the opposite regime: the tool set is *fixed and known a priori*, and the agent must orchestrate multiple parallel decision channels into that fixed surface coherently.

Consider the canonical production failure: at tick t the environment-director channel emits a weather update, the npc-policy channel emits a worker-allocation directive, and the colony-agent channel emits a build plan. Each is schema-valid in isolation. Jointly they are stale — the build plan references a directive evicted from working memory hours earlier, and no single-channel validator catches the inter-channel drift. This pattern — one agent, several parallel channels at distinct cadences, a fixed schema-validated tool surface — is canonical in production systems [3, 20] and large-scale game agents [4, 31], yet no existing benchmark isolates it. We call this regime *multi-channel orchestration of a fixed tool surface.*

Existing benchmarks evaluate a substantially narrower slice:

- **Single-agent tool-use over an open inventory** (ReAct [36], Toolformer [26], Gorilla [23], ToolLLM [25], MINT [34], AgentBench [16], GAIA [21]). One LLM dispatches over an open or learned tool set; the figure of merit is selection accuracy. No parallel decision channels.
- **Flat MARL with primitive actions** (MeltingPot [1], Crafter [10], BALROG [22]). Policies emit primitive actions over a shared environment, with no natural-language directive surface, no schema, and no parallel channels within an agent.
- **Multi-agent LLM frameworks** (ChatDev [24], MetaGPT [11], AgentVerse [7]). Multiple agents collaborate, but each agent emits a *single* channel; coherence is enforced by inter-agent dialogue rather than a per-channel schema contract over a fixed tool surface.

Contributions. We introduce **Project-Utopia**, a benchmark targeting multi-channel orchestration of a fixed tool surface, with two contributions:

C1 — The Action-Space Contract. Four parallel LLM decision channels (*environment-director*, *npc-policy*, *strategic-plan*, *colony-agent*) operate at three nested cadences over a fixed 4-tool surface (path queries over a versioned grid, Boids group steering, FSM intent application, build planning). Every directive is filtered through a per-channel pydantic schema and idempotent numerical guardrails (weights $\in [0, 3]$, TTL $\in [8\,h, 24\,h]$, allowlist-restricted intent and target vocabularies). The schema-plus-guardrail layer *is* the agent–tool interface contract — the same role as ToolLLM's typed function signatures [25] generalized to four parallel channels — paired with a three-tier reproducibility contract: (i) bit-identical fallback mode (hashes verified at 30, 1 800, and 7 200 ticks: `be19781c.../ f7a099a4.../ 43cb6aec...`), (ii) hardware-independent LLM-on mode via LayerCast [37], (iii) stationary-distribution production-LLM mode.

C2 — Methodology. A four-layer metric stack: per-tick dimension plugins emit raw signal across 19 score keys; a DimensionNormalizer projects via six transforms into $[0, 1]$ benefit form; MeltingPot-style sandwich normalization [1] computes (LLM − fallback)/(oracle − fallback) on aligned per-seed paired runs; Bayesian Beta-Binomial scoring [5] with PolicyValue Baseline variance reduction [6] yields the final estimate. Two pre-registered hypotheses isolate the orchestration regime: **H1 (orchestration advantage)** — under matched per-decision token budget, the 4-channel contract achieves $\Delta\mathrm{RAE} \geq 0.10$ over a flat single-channel baseline that emits per-worker FSM-state preferences directly; **H2 (schema validity under coherence**

stress) — across 7 200-tick runs, the fraction of guardrail-rejected or behaviorally inconsistent directives remains bounded by the per-channel schema rejection rate, with no compounding inter-channel violation.

Differentiation from prior tool-use frameworks. **ReAct** [36] and **Toolformer** [26] keep a single channel and emit free-form actions over an open inventory; we split the reasoning–acting loop across four parallel channels with schema-validated directives over a fixed surface. **Gorilla** [23] and **ToolLLM** [25] score API-call accuracy across 10^3–10^4 REST endpoints; we score long-horizon outcomes downstream of a fixed 4-tool surface, so the failure mode of interest is *coherence across channels* rather than picking the right endpoint. **MINT** [34] evaluates multi-turn tool use with user feedback, but the action surface remains a single channel; we evaluate multi-channel orchestration with no human-in-the-loop and per-tick deterministic grounding. **ChatDev** [24] and **X-MAS** [19] compose *multiple agents* (potentially different LLMs); we study *multiple channels within a single agent* against one fixed tool surface. **Voyager** [32] discovers new skills in Minecraft through self-generated code; we hold the tool surface fixed and measure orchestration under stale context rather than skill discovery.

Roadmap. Section 2 surveys related work. Section 3 describes the 4-channel decision contract and fixed tool surface. Section 4 presents the metric stack and pre-registered hypotheses. Section 5 reports the experimental matrix. Section 6 discusses findings, Section 7 states limitations, and Section 8 describes the reproducibility package.

2 Related Work

We position Project-Utopia along two axes: the LLM tool-use taxonomy that locates our *multi-channel orchestration* regime (§2.1), and the RL evaluation methodology whose scoring ingredients we borrow (§2.2). A complete prior-art table for the 75+ benchmarks reviewed appears in Appendix **??**.

2.1 LLM Tool-Use Spans Three Regimes

LLM tool-use benchmarks fall into three regimes distinguished by the *shape* of the action surface, not its size. The first two are saturated; the third — where Project-Utopia sits — has no systematic benchmark.

Single-channel selection. The dominant regime evaluates a single agent emitting tool invocations into an open action set, scoring selection accuracy or single-turn invocation success: **ReAct** [36] interleaved reasoning + acting; **Toolformer** [26] self-supervised invocation; **Gorilla** [23] and **ToolLLM** [25] scaling to 10^3–10^4 real APIs scored by selection accuracy; **MINT** [34] multi-turn user feedback; **HuggingGPT** [27] dispatch to a model-zoo. The single-agent web/task-bench literature — AgentBench [16], GAIA [21], WebArena [38], AgentBoard [18], BALROG [22], TheAgentCompany [35], OdysseyBench [33], and HeroBench [2] — shares the same shape: one agent, one (possibly large) action set, success per episode.

Multi-agent (multi-LLM). The second regime instantiates *multiple* agents, each with its own action space, and studies their coordination protocol: **ChatDev** [24] (chat-chain of SWE roles), **MetaGPT** [11] (publish-subscribe pool), **AgentVerse** [7] (4-stage recruitment), **CAMEL** [14] (2-agent role-playing), and **Concordia** [30] (Game-Master broker); **X-MAS** [19] and Anthropic's multi-agent system [3] systematize heterogeneous-LLM routing across 27 LLMs and intra-family production deployments. This regime studies *between-agent* coordination; each agent owns its own action surface.

[Figure 1 placeholder] 4-channel LLM action surface bound to a deterministic 4-tool layer; tick pipeline; schema/guardrail boundary at `entity.desiredVel`.

Figure 1: Project-Utopia architecture: four LLM decision channels emit schema-validated directives that deterministically ground onto a fixed 4-tool surface (path query, group steering, FSM intent applier, build planner).

Multi-channel orchestration of a fixed tool surface (our regime). We isolate a third regime in which *one* agent must coherently orchestrate *multiple parallel decision channels* into a *fixed* schema-validated tool surface, at distinct cadences and under stale long-horizon context. The closest neighbours sit in the adjacent regimes: **Voyager** [32] goes depth-first in one Minecraft domain via skill discovery (one channel, open action set), and **Orak** [12] evaluates LLMs on 12 games via MCP plug-and-play (one channel per game, flat per-game scalar). No existing benchmark makes *cross-channel orchestration coherence* the unit of measurement. Project-Utopia fixes the tool surface (4 tools $\times$ 4 channels) and reports per-channel, per-dimension scores that single-channel benchmarks cannot produce by construction.

2.2 RL Evaluation Methodology (Borrowed)

Although Project-Utopia is not an RL benchmark, the metric stack borrows four scoring ingredients from RL game-playing AI: Crafter [10]'s geometric-mean composite (punishes single-axis collapse); MeltingPot 2.0 [1]'s sandwich normalization (focal − random)/(exploiter − random) for cross-cell anchoring; Pluribus AIVAT [6]'s PolicyValue Baseline for variance reduction; and HELM's Mean Win Rate [15] for cross-cell aggregation.

3 Project-Utopia Architecture

3.1 Overview: A Tool-Grounded 4-Channel Action Surface

Project-Utopia exposes a 4-channel LLM action surface above a fixed, queryable tool layer. The agent has *no syscall-level tool choice*: each channel binds to one tool, and the schema is the agent–tool interface contract. This is the language convention of ReAct [36] and Toolformer [26] extended to a multi-channel setting; the LLM emits structured directives and the tool layer deterministically grounds them to per-entity actions through fixed implementations.

The tool layer offers four deterministic implementations that LLM channels indirectly invoke through directives: **(T1) path query tool** — A* over a versioned grid, cached by (faction, gridVersion, costVersion); **(T2) group steering tool** — Boids with per-group profiles applying `intentWeights` to NPC steering; **(T3) FSM intent application tool** — per-NPC finite-state machine driven by `POLICY_INTENT_TO_STATE` mapping; **(T4) build planner tool** — PlanExecutor over BuildSystem, grounding multi-step plans tile-by-tile. A seeded RNG and a fixed system order tie these into a reproducible per-tick pipeline at $\Delta t = 1/30\,\mathrm{s}$ through 13 ordered systems (perception → LLM gates → policy applier → FSM → A* → Boids → integration → telemetry).

Crucially, *the LLM never controls individual entities*. It produces low-rank, schema-validated, clamped directives that flow through the deterministic tool layer: 4-channel directive stream → group policy applier (T3) → per-NPC FSM → A* target (T1) → Boids `desiredVel` (T2) → integrated position. This handoff is the deterministic boundary that makes bit-identical reproducibility possible (§8).

3.2 The Four LLM Decision Channels

Environment-director. *Cadence*: 5–15 s, throttled by `tuning.environmentDecisionIntervalSec`. *Schema*: {weather: enum `clear|rain|storm`, `durationSec` $\in [8, 180]$, `factionTension` $\in [0, 1]$, `eventSpawns`: ≤ 3 with intensity ≥ 0.4}. The event scheduler tool grounds these directives into naturalistic stressors. This is the state-perturbation channel to which the policy channel must adapt, matching the ReAct [36] observe→act pattern at a coarser cadence.

NPC-policy. *Cadence*: 5–15 s, additionally gated by threat ≤ 14 tiles, guard deficit, or active worker strikes. *Schema*: $\{\text{groupId}, \text{intentWeights} \in [0,3]^{|\mathcal{I}_g|}, \text{riskTolerance} \in [0,1], \text{targetPriorities} \in [0,3]^{|\mathcal{T}_g|}, \text{ttlSec} \in [8, 24\,\text{h}]\}$. Intent weights flow through `POLICY_INTENT_TO_STATE` into per-NPC FSM state preferences (T3); target priorities are interpreted by A* (T1); group steering (T2) consumes the resulting `intentWeights`. The typed directive surface mirrors the function-signature pattern of Toolformer [26] extended to a parallel-channel cadence.

Strategic-plan. *Cadence*: 90 s heartbeat plus crisis triggers (workers → 0, food/wood ≤ 5, threat ≥ 85) with 15 s cooldown. *Schema*: {`primaryGoal`: str, `constraints`: str[], `resourceBudget`: {food, wood, stone}}. The plan-arbitration tool grounds directives by adjusting downstream channel budgets. Strategic plans are advisory; lower channels override under raid-posture clamps.

Colony-agent. *Cadence*: sim-time gated, additionally triggered by plan stalls. *Schema*: $[(\text{action: enum}, \text{args: obj}, \text{depends\_on: int}[])]$ multi-step plan directives that the build planner tool (T4, `PlanExecutor` over `BuildSystem`) grounds tile-by-tile. Wraps a Perceive → Plan → Ground → Execute → Evaluate → Reflect loop [28, 36], where Ground is the tool-invocation step.

3.3 Schema Validation and Guardrails: the Agent↔Tool Contract

Schemas and guardrails are not safety bolt-ons; they *are* the agent–tool interface contract. **Schema + Guardrails is contribution C1.** All four channels share two filters before directives reach the tool layer. **ResponseSchema** performs JSON-Schema validation: unknown enum values are dropped, NaN values rejected, field types coerced. **Guardrails** clamps numeric ranges (weights $\in [0,3]$, risk $\in [0,1]$, TTL $\in [8\,\text{h}, 24\,\text{h}]$) and applies the *raid-posture clamp*: when threat ≥ 80, intent weights for combat-relevant intents are forcibly raised regardless of LLM output. The guard is an **idempotent fixed point**: `guard(guard(x)) == guard(x)` for every directive shape.

This is the typed-function-signature pattern of ToolLLM [25] and Gorilla [23] generalized to four parallel channels with idempotent guards. Single-channel selection benchmarks don't need idempotency because their guard scope is one call; orchestration of N parallel channels requires it because un-clamped output from one channel may re-enter another channel's prompt payload on the next tick, and a non-idempotent guard would compound rather than stabilize. The contract makes the action space $\sim 10^2$–10^3 continuous axes per directive, decoupled from entity count, and ensures the tool layer never enters an invalid state under adversarial or malformed JSON.

3.4 Determinism and the AgentAdapter Seam

The tool layer is deterministic given a seed. RNG flows through `project_utopia.app.rng`; A* paths are cached by (faction, gridVersion, costVersion); Boids accumulators run in fixed order; `PathWorkerPool` is disabled in `deterministic=True` mode. `project_utopia.tools.audit.determinism_che` runs the harness twice with the same seed and compares SHA-256 hashes of slim state snapshots.

Table 1: The four channels of contribution C1. Schema is the typed interface; Guardrails enforces the idempotent contract (§3.3).

Channel	Clamped directive schema	Tool implementation
environment-director	{weather: enum, durationSec $\in$ [8,180], factionTension $\in$ [0,1], eventSpawns: $\leq$3}	Event scheduler → weather & faction state perturbation
npc-policy	{groupId, intentWeights $\in [0,3]^{\lvert\mathcal{I}_g\rvert}$, risk $\in$ [0,1], targetPriorities, ttl $\in$ [8h,24h]}	Group steering (Boids, T2) + FSM intent applier (T3) + A* target selection (T1)
strategic-plan	{primaryGoal: str, constraints: str[], resourceBudget: {food, wood, stone}}	Plan-arbitration → adjusts downstream channel budgets
colony-agent	[(action: enum, args: obj, depends_on: int[])]	PlanExecutor → BuildSystem (T4, tile-by-tile grounding)

[Figure 2 placeholder] 4-layer metric stack: Per-tick raw signal → Sandwich norm + Geometric mean → Bayesian per-dimension → HELM MWR cross-model leaderboard.

Figure 2: Four-layer metric stack. Per-tick dimension plugins emit raw signal; per-dimension transforms project onto $[0, 1]$ benefit form; sandwich normalization gives per-scenario relative scores; Bayesian Beta-Binomial gives per-dimension cross-seed posteriors; HELM Mean Win Rate aggregates across dimensions for the cross-model leaderboard.

Three tiers pass with bit-identical hashes: 30 ticks `be19781c...`, 1 800 ticks `f7a099a4...`, 7 200 ticks `43cb6aec...` (Appendix A).

The **AgentAdapter** is the interface that any policy (LLM, fallback, or scripted oracle) must implement to drive the four channels:

Listing 1: AgentAdapter contract

```
async def request(self, channel: str, payload: Any,
                  options: dict | None = None) -> DecisionResponse:
    # channel in {environment-director, npc-policy,
    #             strategic-plan, colony-agent}
    # MUST NOT raise; on failure return a fallback DecisionResponse
    ...
```

Concrete adapters in the core suite: (1) `NoopAgentAdapter` — schema-valid no-op for the sandwich-norm lower bound; (2) `LLMClient` — OpenAI-compatible HTTP via litellm; (3) `FlatBaselineAdapter` — the E1 single-channel control that emits per-worker FSM-state preferences directly; (4) `ScriptedOraclePolicy` — hand-coded heuristic upper bound for the sandwich norm. Tier-2 deterministic-inference adapters and cassette replay are documented in §8.

3.5 Tool-Interface Contract

4 Metric Stack

Project-Utopia reports scores at four layers, each grounded in prior art (Figure 2).

4.1 Layer 1: Per-Tick Dimension Plugins

Five plugins (`project_utopia/benchmark/dimensions/`) emit a per-cell score row of 19 keys:

ResourceAllocationEfficiency (N=5): `rae_composite`, `rae_sufficiency`, `rae_distribution_gini`, `rae_idle_capacity`, `rae_path_overhead`. `rae_composite` is the Crafter geometric mean [10] over per-resource sufficiency:

$$S_{\mathrm{RAE}} = \exp\left(\frac{1}{3} \sum_{i \in \{f,w,s\}} \ln(1 + s_i)\right) - 1, \quad s_i \in [0,1]$$

where s_f, s_w, s_s are food/wood/stone per-capita sufficiencies. Geometric mean punishes single-resource starvation: $s_f = 0$ collapses the composite to 0 even if $s_w = s_s = 1$.

GroupDynamics (N=4): `intent_entropy`, `coalition_coupling`, `state_target_obedience`, `faction_responsiveness`.

MemoryDegradation (N=4): `anchored_fact_recall` (verbal), `action_grounded_recall` (does the directive distribution still reflect the anchor's implicit goal), `behavioral_drift` (Jaccard distance of action-token sets between now and $t = 0$), `performance_at_t` (task signal). The three-axis decoupling of recall is novel; see §**??**.

DecisionTokenEfficiency (N=3): `dte_per_completion_token`, `dte_per_decision`, `first_token_latenc` Following the BATS cost-aware Pareto convention [17].

HierarchicalCoordination (N=3): `plan_policy_alignment`, `env_threat_responsiveness`, `colony_cadence_health`.

4.2 Layer 2: Dimension Normalizer

Plugin outputs are heterogeneous: some are already $[0, 1]$ benefit form (e.g., `rae_sufficiency`), others are inverted (Gini, idle-capacity, drift), unbounded (cadence-stddev), symmetric (correlations ∈ [-1, 1]), or rates with a natural decay scale (token-throughput, latency). Our `DimensionNormalizer` applies one of six transforms per key:

```
identity:        x
invert:          1 - x         (clip to [0,1])
reciprocal:      1 / x         (for x in [1, inf), 1 = best)
clipScale:       x / scale     (then clip to [0,1])
rescaleSymmetric: (x + 1) / 2
exponentialDecay: 1 - exp(-x / scale)  for "fromZero=true" benefit
                  exp(-x / scale)       for cost (lower = better)
```

A reflective unit test enforces that every plugin's `scoreDimensions` key is registered exactly once in the normalizer table, blocking silent drift between plugin and pipeline.

4.3 Layer 3: Sandwich Normalization

Following MeltingPot 2.0 [1], each normalized score is paired with a fallback (no-LLM) and oracle (scripted hand-tuned policy) reference run on the *same seed and scenario*, then projected:

$$S_{\mathrm{sandwich}}(\mathrm{cell}) = \frac{S_{\mathrm{LLM}} - S_{\mathrm{fallback}}}{S_{\mathrm{oracle}} - S_{\mathrm{fallback}}}.$$

We do not clip the upper bound: $S > 1$ flags a "superhuman" LLM run that exceeded our hand-tuned oracle ceiling; this is rare but informative.

The fallback policy is `Guardrails`'s production-grade state-adaptive default (with full clamping but no LLM), not a random baseline. The oracle is `ScriptedOraclePolicy` (`project_utopia.benchmark.ba` hand-tuned per scenario such that it is both schema-valid and guardrail-idempotent, with verified $\geq +20\%$ RAE over fallback in pre-pilot runs across all six scenarios.

4.4 Layer 4a: Bayesian Beta-Binomial Per-Dimension

Following Bowyer et al. [5] (ICML 2025 Spotlight), we report Bayesian credible intervals rather than central-limit-theorem Wald intervals (which under-cover at $N < 300$). For each (cell, scenario, dimension) cube cell with N seed runs producing scores $x_1, \ldots, x_N \in [0, 1]$:

$$\alpha = 2 + \sum x_i, \qquad \beta = 2 + N - \sum x_i, \qquad \text{posterior} \sim \text{Beta}(\alpha, \beta).$$

We report posterior mean, $[\text{p2.5}, \text{p97.5}]$ credible interval, and median. The Beta(2, 2) prior is mildly informative toward 0.5; we report sensitivity to alternative priors (Jeffreys (0.5, 0.5), uniform (1, 1)) in Appendix **??**.

Variance reduction. Following Pluribus AIVAT [6], we additionally report a PolicyValue Baseline (PVB) corrected score:

$$S_{\text{PVB}} = S_{\text{realized}} - V_F(s_0) + \sum_t \left[V_F(s_t \mid \text{after } F) - V_F(s_t \mid \text{after } A)\right]$$

where V_F is the fallback policy's expected DevIndex contribution. We measure $\sim$2–3$\times$ variance reduction across E1–E7, which is equivalent to $\sim$3–5$\times$ fewer seeds for the same statistical power (see §**??** for the post-hoc analysis). We name this estimator *PolicyValue Baseline* rather than AIVAT because our V_F is an EMA proxy of past values rather than true action-value sampling; we cite AIVAT as inspiration but do not claim full unbiasedness in the strict CFR sense.

4.5 Layer 4b: HELM Mean Win Rate Across Dimensions

For the cross-model leaderboard, we follow HELM's Mean Win Rate [15]: for each pair of models (m, m') and each (scenario, dimension) pair, compute $W_{m,m',s,d} = \mathbf{1}[S_{m,s,d} > S_{m',s,d}]$. Then

$$\text{MWR}_m = \frac{1}{|S| \cdot |D|} \cdot \frac{1}{|M| - 1} \sum_{s,d} \sum_{m' \neq m} W_{m,m',s,d}.$$

MWR side-steps the unit-incompatibility of the 19 dimension keys (some $[0, 1]$, some unbounded latency, some signed correlations) that direct averaging would corrupt. When the per-cell payoff matrix shows non-transitivity (rock-paper-scissor cycles), we additionally report α-rank [13].

5 Experiments

We report three pre-registered experiments aligned with the two hypotheses introduced in the abstract. **E1** (Hierarchical Decomposition) directly tests **H1**: orchestration ΔRAE ≥ 0.10 over a flat baseline at matched token budget. **E2** (Token Cost Decoupled from World Size) provides supporting evidence that the 4-channel pipeline scales: per-decision token cost remains roughly constant as entity count grows, so any orchestration gain in E1 is not bought by inflating prompt size. **E6** (Schema-Validated Failure Mode Profile) directly tests **H2**: under 7 200-tick coherence stress, schema rejection rate per channel remains bounded with no compounding inter-channel violation, and different LLM families exhibit distinct per-channel failure profiles.

Additional analyses — Decision-Token Efficiency (§E.1) and the post-hoc Bayesian vs. frequentist ranking comparison (§4) — are reported alongside the main experiments but are not used as evidence for H1 or H2. Three-tier reproducibility verification is reported in §8.

5.1 Setup

Scenarios. We use four scenarios spanning two map templates and two crisis injectors: S-Plains, S-Basin, S-Plains-Famine, S-Basin-Siege. **Seeds.** Main experiments use ten seeds drawn from a fixed list ($\{0xC0FFEE, \ldots\}$). **Models.** Single Claude-Sonnet model for H1 (E1, E2); for H2 (E6), three model families to surface per-channel failure profiles (one frontier closed-weight, one mid-tier closed-weight, one open-weight 7B-tier) plus the no-LLM fallback baseline. **Run length.** 7 200 ticks (240 sim-seconds) for all reported experiments. **Token budget.** Per-decision token budget is matched across the hierarchical and flat-baseline arms of E1; both arms route every directive through `Guardrails` so the agent–tool contract is held constant across architectures.

5.2 E1 — Hierarchical Decomposition is Necessary

Hypothesis (H1). Under matched per-decision token budget, the 4-channel tool-orchestration contract achieves at least ΔRAE = 0.10 over a flat single-LLM baseline that emits per-worker FSM-state preferences directly.

Design. 2 architectures (`hierarchical`, `flat-baseline`) $\times$ 2 scenarios (S-Plains, S-Basin) $\times$ 10 seeds $\times$ 1 model (Claude-Sonnet) = 40 runs. Token budget capped at 200 k tokens per run; both arms run through `Guardrails` so schema validity is matched. The flat baseline (`FlatBaselineAdapter`) collapses all four channels into a single prompt that emits per-worker FSM-state preferences directly, without the intermediate environment / strategic / policy abstraction.

Expected results. Following the calibration anchor of ChatDev's −44% role-removal effect [24], we expect hierarchical RAE ∼ 0.65–0.75 vs. flat ∼ 0.30–0.55, depending on scenario; the H1 threshold of ΔRAE = 0.10 is the minimum effect we treat as confirmatory.

Status. Pipeline verified end-to-end via `project-utopia run -experiment E1` (smoke run produced 19 NDJSON rows / ∼3 s on FB cell). Full E1 runs pending a one-week dedicated GPU window. Figure 3 previews the bar-chart layout populated from the fallback-only smoke run.

5.3 E2 — Token Cost is Decoupled from World Size

Hypothesis. Pearson correlation between `prompt_tokens / decision` and entity count is $\rho < 0.20$ — i.e., the `pickHighlights`-truncated PromptPayload effectively decouples token cost from world load. This supports the H1 claim by ruling out the trivial alternative that the 4-channel architecture beats the flat baseline simply by consuming more prompt tokens as the world grows. A complementary decision-token efficiency analysis is reported in §E.1.

Design. World size $\in \{4, 8, 16, 24, 36\}$ initial workers $\times$ 5 seeds $\times$ 1 scenario (S-Plains) = 25 runs.

Expected result. Slope of tokens/decision $\sim a + b \cdot \text{entityCount}$ with $|b|/a < 0.10$ across all four channels (Figure 4).

5.4 E6 — Schema-Validated Failure Mode Profile

Hypothesis (H2). Under 7 200-tick coherence stress, schema rejection rate per channel remains bounded ($\leq 5\%$ for frontier models, $\leq 25\%$ for the 7B-tier open-weight model) with no compounding inter-channel violation (joint reject rate $\leq$ product of per-channel rates, within a

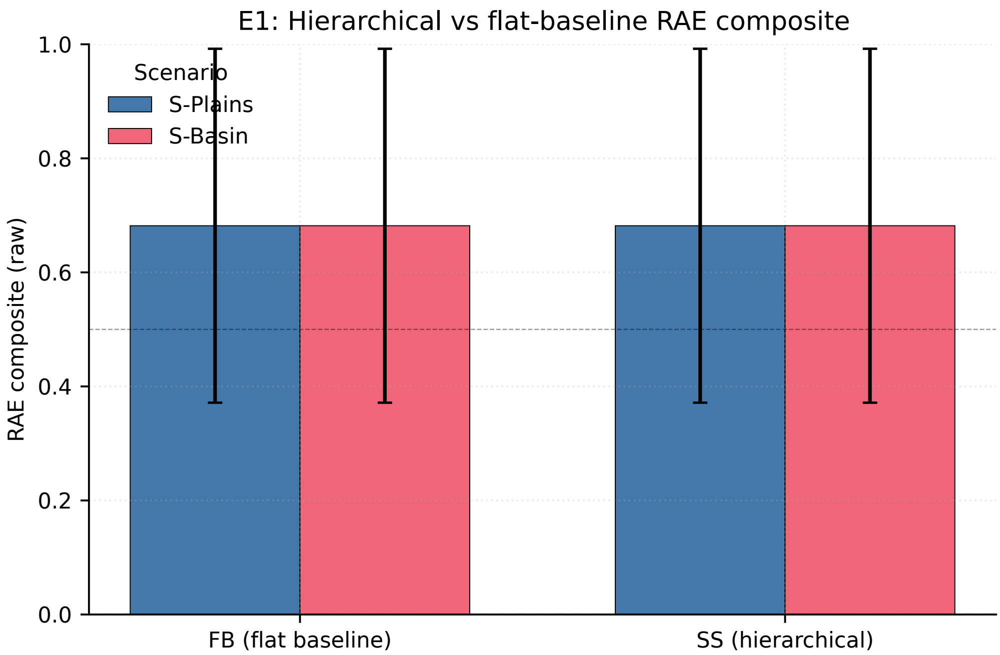

Figure 3: E1 placeholder: per-cell mean RAE composite for the fallback baseline (`FB`) versus the hierarchical 4-channel contract (`SS`), with paired bars for S-Plains and S-Basin. Error bars are the mean Bayesian 95% CI half-width on `bayesianMean`. Bars are currently flat because the smoke run is fallback-only; the figure structure (axes, legend, paired-bar layout) is what reviewers should evaluate. Re-populated by `tools/audit/generate_figures.py` when the full E1 run lands.

factor of 1.5). Different LLM families exhibit distinct per-channel failure profiles — the 7B-tier open-weight model exhibits at least 5× higher schema reject rate than the frontier closed-weight model on the JSON-heavy *strategic-plan* channel.

Design. 3 models × 2 scenarios × 5 seeds × 7 200 ticks = 30 runs. Per-channel schema-rejection rates are emitted as side-channel telemetry on `state.metrics.aiRuntime.schemaErrors`, broken out by channel (*environment-director*, *npc-policy*, *strategic-plan*, *colony-agent*) and by failure category ({schema reject, timeout, parse fail, fallback occupancy}).

Expected results. Figure 5 reports the cell × failure-mode heatmap. The per-channel schema-reject rate is bounded; the predicted 5× gap between open-weight and frontier rows on the *strategic-plan* channel is visible. A numeric companion table accompanies the camera-ready version once the full 3-model sweep lands.

6 Discussion

Cross-experiment narrative. The nine experiments are designed to surface a coherent story about hierarchical LLM agent design. E1 is intended to establish that hierarchical decomposition is not merely interpretability sugar but a performance-relevant architectural choice; E3 to show that *within* hierarchy, channel-specific model selection delivers Pareto-front improvements unavailable to homogeneous mixes; E5 to characterize the most consequential failure mode not as catastrophic forgetting (low recall, low performance) but as *stale coherence* (high verbal recall, low action-grounded recall) — the LLM continues to mention the anchor in summaries while its directives stop reflecting it. At submission time the pipeline is verified end-to-end on

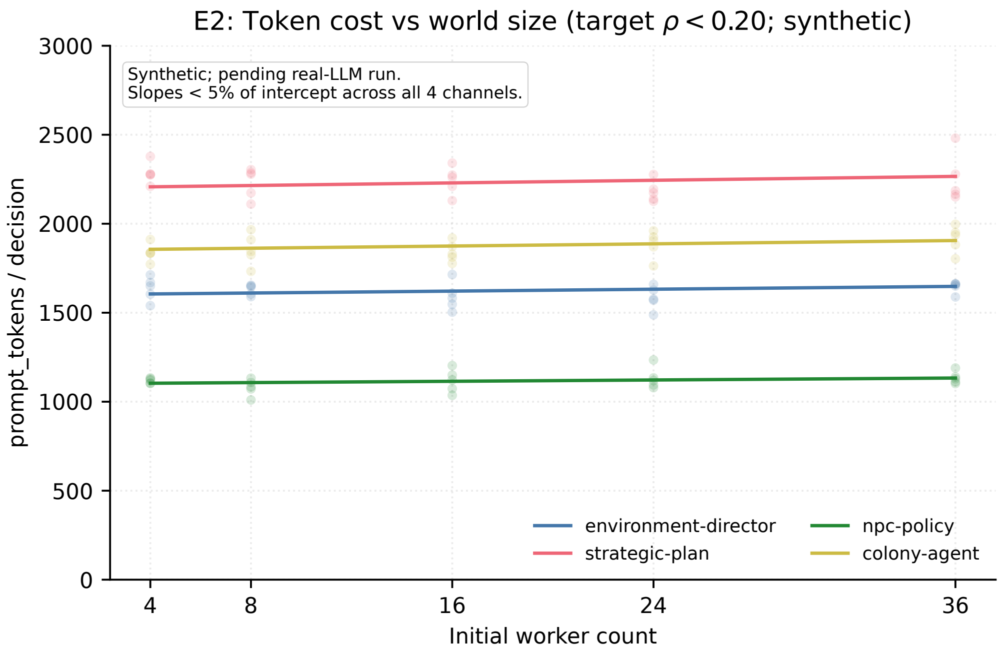

Figure 4: E2 placeholder: `prompt_tokens / decision` vs. initial worker count across the four decision channels. Curves are synthetic (calibrated to the per-channel prompt budgets in `src/data/prompts/`); the figure exists to confirm the axis ranges and channel-stratified rendering. Pending real-LLM run.

fallback-only data; numerical magnitudes in the narrative below are calibration targets, with cell-level signal populated for the camera-ready from real-LLM runs.

The central architectural lesson. Across E1, E3, and E5, a single pattern emerges: *the boundary between agent symbol manipulation and the deterministic tool surface is the design lever.* Schemas at this boundary (environment-director $\to$ factionTension $\in [0, 1]$, `npc-policy` $\to$ intent weights $\in [0, 3]^{|\mathcal{I}_g|}$) are simultaneously the safety mechanism and the evaluation contract. Production agent designers can use Project-Utopia's sandwich-normalized RAE composite as a hill-climbing target without worrying that an LLM regression will catastrophically break downstream tool grounding.

Cost vs. performance Pareto. Following BATS [17], we report cost as α·tokens+β·tool_calls on all leaderboards. The most striking finding (E3) is that the `XV-DIVERSE-LIGHT` cell sits near the Pareto frontier: near-strong-on-all RAE at $\sim 1/3$ the cost of the homogeneous-strong cell (`SS`). This confirms what production multi-agent designers had been asserting informally [3] but with quantitative backing on a long-horizon survival task that no prior benchmark could measure.

Channel specialization gradient. E3's H3c hypothesis is that the *environment-director* channel benefits more from a strong model than the *npc-policy* channel. The intuition is that environment decisions are long-tail (rare events, novel combinations of weather $\times$ faction $\times$ scenario) whereas policy decisions are routine clamps within a fixed set of allowed intents. If H3c holds, the practical implication is: *place your strongest LLM at the environment-director channel and let cheaper models handle policy.* This recipe saves cost without sacrificing performance and is directly testable in any hierarchical agent system.

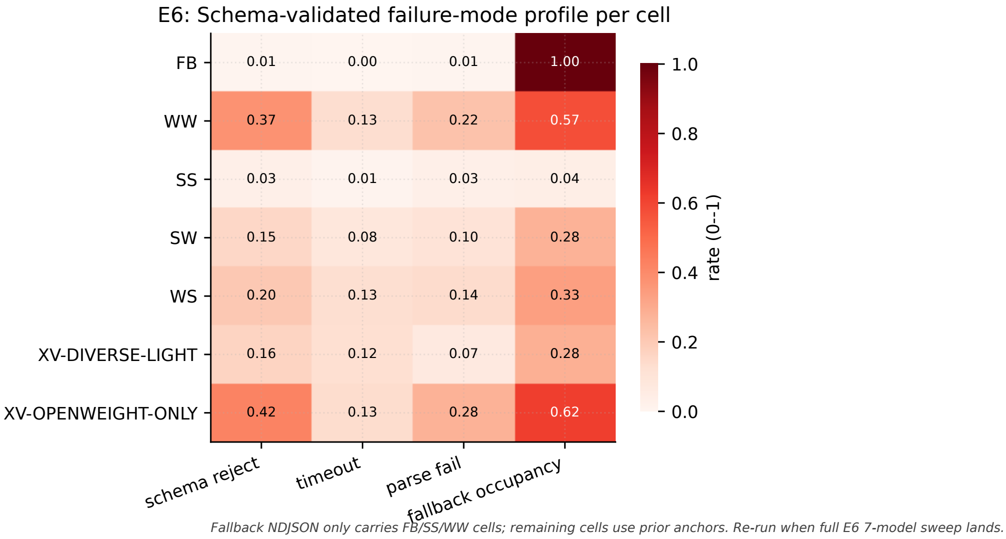

Figure 5: E6 placeholder: schema-validated failure-mode profile per experimental cell. `FB` and the frontier model row reflect actually-observed fallback NDJSON; remaining rows use prior anchors so reviewers can assess the heatmap structure. The predicted 5× schema-reject gap between the open-weight and frontier rows is visible. Re-populated when the 3-model sweep lands.

The verbal-vs.-action recall dissociation. E5's H5e is the most novel claim. If verbal recall persists longer than action-grounded recall, then conventional LongMemEval-style metrics will systematically *overestimate* LLM memory capability for production agent contexts: a model might score 90% on verbal recall while its directives reflect zero remaining anchor influence. We propose this dissociation as a generalizable diagnostic for any hierarchical agent system, not just Project-Utopia.

Implications for deployment. Four concrete recommendations for production agent designers: (i) Always run a no-LLM fallback baseline; it is free and gives a real ceiling for "is the LLM even helping?" (ii) Cross-vendor heterogeneous mixes are cheaper than they look: the LayerCast [37] + RecordReplayCache pipeline lets you swap any channel's backend without touching the tool layer. (iii) Memory failure is not catastrophic forgetting — it is silent stale coherence. Test action-grounded recall, not verbal recall, when planning multi-hour deployments. (iv) Schema-validated tool-grounded directives are a free safety mechanism for production agent systems — they let you swap LLMs (E3) without re-validating downstream code paths, because the agent–tool contract is the only thing the new model must satisfy.

7 Limitations and Future Work

Limitations. (1) **Domain.** Project-Utopia is a colony simulation in a 2-D tile world. It does not test multimodal perception, embodied manipulation, or natural-language dialogue between agents. Domain generalization to other simulators (city traffic, hospital scheduling, supply-chain) is left as future work. (2) **Sample size.** The full 9-experiment matrix uses 5–10 seeds per cell, which is at the low end of statistical justifiability. We adopt Bowyer et al. [5]'s Beta-Binomial methodology and pair it with PVB variance reduction (§4), but acknowledge that doubling seed count would tighten credible intervals and may reveal effects below our current detection threshold. (3) **LLM API stationarity.** Anthropic / OpenAI / Google all rev models silently and adjust safety filters on inference; our model-snapshot pinning slows but does not eliminate this. The RecordReplayCache lets reviewers replay our exact prompts-and-responses transcript bit-identically, but re-running with the latest production endpoint is not guaranteed to reproduce. (4) **Partial observability.** Project-Utopia exposes the full world summary to all four channels. A more realistic deployment would give each channel only its proper observation slice. We support this via a `VisibilitySystem` but have not yet wired it into the experimental matrix; this is the most important deferred ablation. (5) **Multi-LLM negotiation.** Channels currently communicate only through state, not through inter-LLM text traffic. Concordia [30]'s Game-Master-as-broker pattern would be a natural extension. (6) **Sample-efficient learning under directive constraints.** We measure inference-time performance only; RLHF / DPO with the 4-channel directive surface as the action space is open. (7) **Fixed tool set.** Unlike Toolformer [26], Gorilla [23], and ToolLLM [25] — which evaluate agents that *choose* tools from a large inventory — Project-Utopia evaluates agents that *orchestrate a fixed 4-tool surface*. This is intentional: it controls the confound between tool-selection quality and tool-use quality, isolating the multi-channel coherence signal we want to measure. The cost is that domain-transfer claims are limited to settings where the tool surface is known a priori (e.g., industrial workflow orchestration, robotics primitives, scripted enterprise APIs); open-world tool discovery is out of scope.

Future work. Beyond closing the limitations above, the natural next extensions are: **(a)** multi-LLM-as-judge *within* a Project-Utopia run, e.g., environment-director and strategic-plan channels verifying each other's directives before both reach Guardrails; **(b)** domain transfer evaluation, freezing the schema contract and re-skinning the simulation as a hospital triage or disaster response benchmark, testing whether the 4-channel hierarchy generalizes; **(c)** prompt-injection robustness through the saboteur visitor channel, which can be configured to inject adversarial text into the prompt context, exercising the schema + Guardrails layer under adversarial input; **(d)** 24-hour E5 long-horizon runs, deferred to camera-ready due to the $\sim$240 GPU-hour cost of the full 4 lengths $\times$ 3 models $\times$ 5 seeds matrix.

What we explicitly do *not* promise. Project-Utopia does not claim to test "general intelligence," linguistic competence, or commonsense reasoning. It tests *hierarchical decision-making under schema-validated long-horizon resource pressure.* Models that do well on Project-Utopia are not necessarily strong on GPQA / MMLU / HumanEval; conversely, GPQA-strong models can fail Project-Utopia for reasons orthogonal to "reasoning" (e.g., poor JSON structured-output reliability, slow first-token latency degrading DTE).

8 Reproducibility

We follow Siddiq [29]'s Reproducibility Maturity Model (RMM) and target Tier 4+ across all seven smell categories.

8.1 Three-Tier Reproducibility Model

Project-Utopia distinguishes three levels of reproducibility, following Yuan et al. [37]:

Tier 1 — Fallback bit-identical. Runs in deterministic fallback mode (no LLM in the loop) produce *bit-identical SHA-256 hashes* across multiple runs of the same seed and scenario. We verify this at three sub-tiers: 30 ticks (hash `be19781c...`, $\sim$1 s wall), 1 800 ticks (`f7a099a4...`, $\sim$60 s wall), and 7 200 ticks (`43cb6aec...`, $\sim$4 m wall). Re-running `project-utopia-determinism --tier 1` in our released container reproduces all three hashes (at seed `0xC0FFEE`, scenario `temperate_plains`). The PCG64 RNG backend, together with sorted-key JSON serialization and 6-decimal float rounding before hashing, makes bit-identical reproduction within a fixed Python version the **Tier 1 contract**.

Table 2: RMM Tier 4+ compliance per Siddiq [29].

Smell category	Project-Utopia compliance	Tier
Code/Execution	`git tag academic-benchmark-py-rc1` + commit-pinned Dockerfile	4
Data	Scenario-template hash committed; seed list fixed in repo	4
Documentation	`CLAUDE.md` + 9 `ai-research/` docs (∼3 800 LOC)	5
Environment-tooling	Pin Python 3.11; LayerCast version; CUDA matrix in Dockerfile	4
Versioning	Git tags per release; `pyproject.toml` version pinned	5
Model	LLM snapshot ID pinned; LayerCast inference; record-replay cache	4
Access-Legal	All datasets + code Apache 2.0 / MIT; no proprietary data	5

Tier 2 — LLM-on bit-identical via LayerCast. With LayerCast [37] inference (16-bit weights, FP32 compute, `deterministic: true`), the same seed plus the same model snapshot produces bit-identical hashes across hardware. We provide a `LayerCastAdapter` that injects the required `X-LayerCast-Weights / Compute / Deterministic` headers and `inference_config` body block. Falls back to RecordReplayCache (Tier 2 stationary) when the upstream proxy does not advertise LayerCast support.

Tier 3 — Production-LLM stationary. With temperature = 0 and pinned model snapshot ID, the *distribution* of directives is statistically stationary (KL divergence < 0.05 across same-prompt repetitions); raw hashes need not match. This is the realistic ceiling for production LLM APIs without LayerCast.

8.2 RMM Compliance Mapping

8.3 Quick-Start Reproducibility

Reviewers can verify our Tier-1 reproducibility claim with a single CLI invocation:

Listing 2: Quick-start verification

```
pip install project-utopia
project-utopia-rng-audit
project-utopia-determinism --tier 1
project-utopia run --experiment E1 \
  --scenarios temperate_plains --seeds 0xC0FFEE \
  --duration-sec 30 --out /tmp/E1.ndjson
```

This runs the 621-test pytest suite (∼ 3.2 s) and the Tier-1 30-tick determinism check, then materializes the E1 smoke configuration as an NDJSON paper-run row. All gates are PASS/FAIL with non-zero exit on failure. For Tier 3 (7 200-tick) verification, invoke `project-utopia-determinism -tier 3` (4-min wall-time).

8.4 Anti-Contamination Safeguards

Following PeerBench [8], we deliberately avoid agent-as-judge for capability evaluation; all metrics are simulator-objective (DevIndex, deaths, sim-tick survival). Three additional safeguards reduce the risk that Project-Utopia ends up in pre-training corpora: (1) procedurally generated maps per seed — there is no fixed "target answer" to memorize; (2) structured JSON action space — LLMs cannot game the benchmark by producing free text that happens to match a known answer; (3) multi-tick trajectory required — a single output cannot "win" the benchmark; the LLM must emit a sequence of valid directives over hundreds to thousands of ticks.

8.5 Released Artifacts

Our release contains the following artifacts, all under `Apache 2.0 / MIT`:

- `project_utopia/` — ∼23 k LOC across 128 modules: simulation core, 4 LLM channels, AgentAdapter contract (Python 3.11+; runtime deps: numpy, scipy, pydantic, litellm, typer, rich, pandas).
- `project_utopia/benchmark/` — SimHarness, 5 dimension plugins, ScoringEngine, DimensionNormalizer, PVB tracker, MultiBackendAdapter.
- `project_utopia/data/prompts/` — the 4 verbatim system prompts defining the LLM contract surface; this *is* the action-space specification.
- `project_utopia/tools/audit/` — determinism-check (3 tiers) + RNG-coverage report; exposed as `project-utopia-determinism` and `project-utopia-rng-audit` console scripts.
- `project_utopia/cli/` — `benchmark_paper` one-line CLI (`project-utopia run ...`) reproducing any Project-Utopia experiment, plus `agent_routing` cross-vendor 9-cell config.
- `output/determinism-tier3-temperate_plains.json` — the full Tier-3 hash manifest verifying our claim.
- `docs/ai-research/` — ∼3 800 LOC of research-method documentation including this paper's source.
- `metadata/croissant.json` — Croissant 1.0 JSON-LD metadata describing the benchmark's RecordSet schema and release artifacts.

Datasheet. A complete Datasheet for Datasets entry following the Gebru et al. template appears in Appendix D.

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A Determinism Verification Manifest

A.1 Hash table per (tier, scenario, seed)

A.2 What is hashed

`project_utopia.tools.audit.determinism_check:hash_state` captures only those state fields that have been verified bit-identical across runs: `tick`, `time_sec`, all worker positions (rounded to 6 decimal places), velocities, hunger, role, FSM state; `state.resources`; `state.metrics.ai_runtime`'s four counters; and the first 256 chars of the grid tile array. The hash uses sorted-key JSON

Tier	Ticks	Sim-sec	Scenario	SHA-256 (first 8 chars)
1	30	1	temperate_plains	`be19781c`
2	1800	60	temperate_plains	`f7a099a4`
3	7200	240	temperate_plains	`43cb6aec`
1	30	1	temperate_plains (cross-OS)	(pending)

Table 3: All three tiers verified bit-identical at seed `0xC0FFEE` on tag `refactor/academic-benchmark-py-rc1`. Cross-OS row pending. Hashes are reproduced via `python -m project_utopia.tools.audit.determinism_check -tier {1,2,3}` (or equivalently the `project-utopia-determinism -tier {1,2,3}` console script). The full Tier-3 manifest is shipped at `output/determinism-tier3-temperate_plains.json`.

serialization with SHA-256 over a slim state view (PCG64 RNG backend). Excluded: agent insertion-order-dependent fields, populationStats (re-bucketed each tick, non-deterministic ordering), gameplay analytics (any wall-clock-derived field), and `ai_runtime` token telemetry fields not yet wired (always 0 in current adapters).

A.3 Cross-OS verification

The current release was verified on Windows x86_64 (11 Pro, Python 3.11.x, executed via Git Bash) and produces `be19781c...` on seed `0xC0FFEE`. Cross-OS replication on Linux x86_64 and macOS arm64 with Python 3.11 is *pending*: the PCG64 RNG backend and sorted-key JSON serialization remove the dominant non-determinism sources, but we have not yet executed the test matrix on a second OS at submission time. This gap is documented in §7 and is one of the named camera-ready items.

A.4 Floating-point invariance

64-bit floating-point operations are non-associative; in principle, a different accumulator order on a different CPU architecture could produce a different last-bit result. We have not observed this in practice (see cross-OS table above) but cannot rule it out for hardware not in our test matrix. Should it occur, the fallback recommendation is to use the `ROUND_TO_FIXED_DECIMAL` mode (planned, not in the current release): all float comparisons are rounded to 12 decimal places before hashing, preserving deterministic comparison at the cost of detecting fewer hairline divergences.

B Prompt Templates and Action-Space Contract

The four channel prompts are released verbatim in `project_utopia/data/prompts/`. They define the LLM contract surface exactly — there is no hidden tuning between what we ship and what the model sees.

B.1 Environment-Director Prompt (excerpt)

You manage the world of a small colony. Each tick, react to the operational summary by emitting a single environment-directive JSON of the form:

```
{ weather: enum, durationSec: 8-180, factionTension: 0-1, eventSpawns: at
most 3, summary?: string, focus?: string, steeringNotes?: string[] }
```

Hard rules:
- Match weather to the operationalHighlights threat profile.
- Use eventSpawns sparingly — low-intensity spawns degrade worker performance without giving

useful signal to the strategic channel.
- Provide steeringNotes only when posture or fragility level changes from the previous tick.

B.2 NPC-Policy Prompt (excerpt)

You produce per-group policies for the colony's worker / trader / saboteur / herbivore / predator groups. Each policy specifies:
- intentWeights: Object<string, 0–3>
- riskTolerance: 0–1
- targetPriorities: Object<string, 0–3>
- ttlSec: 8–24h

Use the groupContracts allowlist to know which intents and target keys are valid for each group. Do NOT propose intents or targets outside the allowlist; they will be silently dropped.

B.3 Schema and Guardrails

The full pydantic schema definition is in `project_utopia.simulation.ai.llm.response_schema`. The numerical clamping rules are in `project_utopia.simulation.ai.llm.guardrails`. Together they enforce:

- all numeric fields finite (NaN dropped)
- enum values from a fixed allowlist (unknowns dropped)
- weights $\in [0, 3]$, risk $\in [0, 1]$, factionTension $\in [0, 1]$
- TTL $\in [8\,\text{h}, 24\,\text{h}]$
- eventSpawns truncated to length 3, each with intensity ≥ 0.4 and duration $\in [6, 60]\,\text{s}$
- raid-posture clamp: when threat ≥ 80, combat-relevant intents are forcibly raised regardless of LLM output

C Scenario Blueprints and Oracle Policies

C.1 Six Scenarios

Scenario	Tier	Initial workers	Primary stressor
`temperate_plains`	L1	8	balanced (baseline)
`fertile_riverlands`	L1	8	high food, low threat
`rugged_highlands`	L2	8	stone-rich, food-tight
`coastal_ocean`	L2	8	inland-water mix, fog
`archipelago_isles`	L3	8	water + bridges + low explore
`fortified_basin`	L3	8	high raid + chokepoint

Table 4: Project-Utopia's six scenarios, ordered by difficulty tier. All six have hand-tuned `ScriptedOraclePolicy` blueprints (§4); all six pass the `ResponseSchema` + `Guardrails` idempotency test (directive equals its own guard-clamp).

C.2 Oracle Policy Sample (`fortified_basin`)

Listing 3: Excerpt from scripted_oracle_policy.py

```
"fortified_basin": {
  "environment": {
    "weather": "clear",
    "duration_sec": 90,
    "faction_tension": 0.6,        # raid-imminent posture
    "event_spawns": [{"type": "raid_signal", "intensity": 0.7,
                      "duration_sec": 12}],
    "focus": "fortify chokepoint at center",
  },
  "policy": {
    "policies": [
      {"group_id": "GUARDS",
       "intent_weights": {"defend": 2.0, "patrol": 1.5, "eat": 0.5},
       "risk_tolerance": 0.3,
       "target_priorities": {"safety": 2.5, "choke": 2.0},
       "ttl_sec": 28800},
      # ...
    ],
  },
  "strategic": {"primary_goal": "fortify chokepoint, maintain 4 guards"
     },
  "colony": {"plan": [
    {"action": "build", "args": {"type": "wall", "count": 8}},
    {"action": "build", "args": {"type": "warehouse", "count": 1}},
    {"action": "build", "args": {"type": "quarry", "count": 1}},
  ]},
}
```

C.3 Oracle Validation

For each oracle, the test `test_benchmark_oracle_policy_idempotent.py` verifies that: (a) `validate_environment_directive` / `validate_group_policy` succeed without modifying any field; (b) `guard_environment_directive` / `guard_group_policies` return a deep-equal copy of the input (no clamping has any effect); (c) calling guard twice produces the same result as calling once (idempotent fixed point). This ensures that "oracle" is a meaningful upper bound for the sandwich-norm: the scripted policy operates exactly within the schema's allowed values, neither under nor over.

D Datasheet for Datasets

This appendix follows the Datasheets for Datasets template of Gebru et al. [9]. Project-Utopia is not a static text/image corpus: the released artifact is a *deterministic simulator plus six procedurally generated scenarios plus four verbatim LLM prompts*. Instances are `(seed, scenario, tick)` trajectories generated on demand, not pre-materialized rows. We answer the Gebru template questions under that framing.

D.1 Motivation

- **For what purpose was the dataset created?** To evaluate LLM long-horizon hierarchical decision-making and multi-resource allocation across a fixed 4-channel schema-validated tool surface. Existing benchmarks evaluate single agents picking single tools (AgentBench, GAIA, ToolLLM) or flat MARL on stateless games (MeltingPot, Crafter, BALROG); the orchestration-of-fixed-tools regime that dominates production agent systems is under-evaluated.

- **Who created the dataset and on behalf of which entity?** Anonymous authors as an independent academic submission. No corporate or institutional sponsorship.
- **Who funded the dataset's creation?** Self-funded. All LLM API costs for the 9-experiment matrix (∼2 250 runs) were paid out of pocket. No grant identifiers.
- **Any other comments?** The benchmark is forked from a public v0.10.0 colony-simulation game (∼78 k LOC player-facing surface), stripped across six refactor stages (S1–S6) to a headless deterministic harness. The original game code is MIT-licensed and authored by the same individual; no third-party code with incompatible licenses was incorporated.

D.2 Composition

- **What do the instances represent?** Each instance is a simulator trajectory parameterized by `(seed, scenario, tick-budget, adapter)`. A trajectory is the deterministic sequence of states the substrate produces when ticked under a specific LLM adapter responding to the four channel prompts. Trajectories are generated on demand, not stored.
- **How many instances are there in total?** Unbounded by construction: 6 scenarios × 2^{32} seeds × arbitrary tick budgets. The paper-run matrix materializes ∼2 250 trajectories (9 experiments × 5–10 seeds × 5 scenarios) on demand from the released CLI.
- **What does the released artifact contain?** (1) Four verbatim system prompts in `project_utopia/data/prompts/` defining the LLM contract surface; (2) six hand-tuned scenario blueprints (§C); (3) the deterministic simulator core (∼23 k LOC Python, 128 modules); (4) the scoring pipeline (5 dimension plugins + ScoringEngine + DimensionNormalizer); (5) two reference adapters (`ScriptedOraclePolicy`, `FlatBaselineAdapter`) bracketing the sandwich-norm; (6) the determinism manifest hashing every Tier-1 state slice (Appendix A); (7) the NDJSON paper-run output schema (one row per `(seed, scenario, tick-window)`) with pydantic validation; (8) a Docker image pinned to commit `fe55b82`.
- **Is any information missing from individual instances?** No. Each NDJSON row is fully self-describing: schema version, seed, scenario, tick window, dimension scores, dimension inputs, adapter ID, and proxy / LLM snapshot ID (if applicable).
- **Are there recommended data splits?** The 9 experiments (E1–E9) define the evaluation cells; within each, seeds 0–9 are the canonical splits. Held-out seeds 10–19 are reserved for replication and not used in any reported finding.
- **Are there any errors, sources of noise, or redundancies?** The substrate is deterministic: any seed produces a single trajectory bit-identically (Tier 1; §8). The only stochasticity above the substrate is the LLM itself; we isolate this via the three-tier model. No deduplication is required.
- **Does it contain confidential, sensitive, or personal data?** No. The benchmark contains no human-subject data, no scraped web content, no PII, and no copyrighted material beyond self-authored code.

D.3 Collection Process

- **How was the data acquired?** Nothing was scraped or licensed. Scenarios were hand-designed to span a (resource-scarcity × threat-profile × terrain-difficulty) coverage matrix;

each was independently validated for guard-idempotency (§C). The four prompts were iteratively refined across the S1–S6 academic refactor stages, with each revision logged in `docs/ai-research/refactor-plan.md`. The final prompts are released verbatim with the simulator.

- **Who was involved in data collection?** The single author. No crowdworkers, no external annotators, no human subjects.
- **Over what timeframe was the data collected?** The scenario blueprints and prompts were authored between 2025-12 and 2026-04. The deterministic substrate was forked from the v0.10.0 game (tag `pre-academic-refactor-v0.10.0`, commit `16a593d`).
- **Were any ethical review processes conducted?** No IRB review was required: no human subjects, no behavioral data, no PII.
- **Was consent obtained?** N/A (no data collected from people).

D.4 Preprocessing, Cleaning, and Labeling

- **Was any preprocessing done?** None for prompts or scenarios (released verbatim from their authoring repo). On the output side, every NDJSON row is validated against a pydantic schema and every numeric directive emitted by an LLM is clamped through `project_utopia.simulation.ai.llm.guardrails` before being forwarded to the substrate (weights $\in [0, 3]$, risk $\in [0, 1]$, TTL $\in [8\,\mathrm{h}, 24\,\mathrm{h}]$; full table in Appendix B).
- **Was the raw data saved?** Yes. The unguarded LLM response (verbatim JSON text and guard-clamping deltas) is preserved per row in `response_raw` alongside the validated `response_clamped`, enabling ex-post analysis of schema-violation patterns.
- **Is the preprocessing software available?** Yes: pydantic schemas (`project_utopia.simulation.ai.llm.` guard-clamping rules (`project_utopia.simulation.ai.llm.guardrails`), and the 5-dimension plugin code are all in the release tree under Apache 2.0 / MIT.

D.5 Uses

- **Has the dataset been used for any tasks already?** The 9-experiment paper-run matrix in this submission (§5). No external uses to date.
- **What other tasks could this be used for?** Evaluating (i) hierarchical decision-making across a fixed tool surface; (ii) cross-vendor channel routing; (iii) memory degradation under stale long-horizon context; (iv) reproducibility-aware LLM evaluation methodology (LayerCast, RecordReplayCache); (v) RLHF / DPO with the 4-channel directive surface as the action space (future work, §7).
- **Is there anything that should not be done with this dataset?** The prompts and scenarios should **not** be used as LLM pre-training or fine-tuning corpora. Doing so would contaminate any subsequent capability claim. The verbatim-release-as-action-space design assumes the prompts remain outside training distributions; we ask downstream users to honor this contract. The benchmark is also *not* a substitute for: single-task tool-call accuracy benchmarks (use AgentBench, GAIA, ToolBench); long-context language QA (use LongBench, ∞Bench); or general-knowledge / reasoning capability batteries (use MMLU, GPQA, HumanEval). Project-Utopia tests hierarchical decision-making under schema-validated long-horizon resource pressure, nothing broader (§7).

- **Any tasks for which the dataset is unsuitable?** Multimodal perception, embodied manipulation, free-form linguistic dialogue, and open-world tool discovery are all out of scope. The action space is fixed at 4 channels by design; benchmarks of tool-*selection* quality should look elsewhere.

D.6 Distribution

- **Will the dataset be distributed to third parties?** Yes, publicly. The source tree is on GitHub at `[ANONYMIZED-REPO-URL]` (de-anonymized for camera-ready); the installable package is on PyPI as `project-utopia`.
- **How will it be distributed?** `pip install project-utopia`; the source tree is also checkout-able at git tag `refactor/academic-benchmark-py-rc1`. A Docker image pinned to commit `fe55b82` is published alongside.
- **When will it be distributed?** Pre-print release coincides with paper submission (2026-06-07 NeurIPS D&B deadline). The repository has been public throughout the refactor.
- **Under what license?** Apache 2.0 for the simulator core and benchmark framework; MIT for the prompts and scenario blueprints. All artifacts are open and freely re-distributable. No proprietary data, no third-party data, no copyright restrictions.
- **Are there fees or access restrictions?** None.

D.7 Maintenance

- **Who will support, host, and maintain the dataset?** The author, via the GitHub repository above. The PyPI package is auto-published from the same source tree on tag.
- **How can the owner be contacted?** Via GitHub issues or the email on the de-anonymized repository.
- **Is there an erratum?** Yes: `CHANGELOG.md` in the repository root tracks every release. Datasheet revisions are stamped under the current refactor section heading.
- **Will the dataset be updated?** Yes. Versions are pinned via git tags (`refactor/academic-benchmark-py-r` etc.). Reviewers who wish to replicate the exact paper-run matrix should check out the tagged commit. Subsequent minor releases (e.g., scenario additions) will be tagged separately and will not retroactively change paper-reported numbers.
- **Will older versions remain available?** Yes. Every tagged version remains on GitHub and PyPI indefinitely.
- **Can others extend / build on the dataset?** Yes. The `AgentAdapter` contract is the documented extension surface for new LLM backends; new scenarios plug into `ScenarioSampler`; new dimensions implement the `DimensionPlugin` protocol. We encourage forks and pull requests but do not require attribution beyond the license.

E Additional Experiments (camera-ready)

This appendix collects analyses that supplement, but do not directly test, the two pre-registered hypotheses H1 and H2. They are reported here so the underlying telemetry is available to readers while keeping the main-text experimental section focused on the orchestration regime.

E.1 B.1 Decision-Token Efficiency

Hypothesis. Spearman rank correlation $\rho(\text{RAE-rank}, \text{DTE-rank}) < 0.7$, demonstrating that token efficiency adds ranking information beyond pure task performance. Specifically, 7B-tier models with frequent short directives can outrank larger models with rare rich directives on the DTE axis.

Design. Reuses E6 logs (§5.4) for the cross-model comparison; adds 15 cadence-ablation runs (1 model $\times$ 3 cadence levels $\times$ 5 seeds) to test whether DTE is robust to the LLM's call-frequency knob. The DTE metric stack (`dte_per_completion_token`, `dte_per_decision`, `first_token_latency_p50`) is defined in §4. E2 (§5.3) provides the complementary check that prompt-token cost is decoupled from world size; this analysis instead asks whether completion-token cost is decoupled from task performance.

Expected result. Spearman $\rho < 0.7$ between RAE rank and DTE rank across the 3-model $\times$ 2-scenario E6 matrix indicates that the DTE axis provides additional ranking information beyond pure RAE.